IDIONG, ESANGUBONG JIMMY

ML/AI Engineering

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PROFESSIONAL SUMMARY

AI Engineer with a Computer Engineering background and two production health AI systems in deployment. Built a clinical diabetes risk classifier (CatBoost, NHANES, custom clinical thresholds, 0.782 diabetes recall) and a medical RAG system (LLaMA-3, Modal, multi-stage retrieval, streaming inference), both end-to-end, from data pipeline to frontend. Seeking ML/AI engineering roles.

PROJECTS

DiaWatch - Diabetes Risk Prediction System

Python, Scikit-learn, CatBoost, Vercel, MLFlow, FastAPI, Docker, Google GenAI, TypeScript, PostgreSQL, HuggingFace, Ray Tune

- Designed a 3-class diabetes risk classifier (Healthy / Prediabetes / Diabetes) on 19,125 NHANES records using CatBoost with MICE imputation and engineered clinical features; hyperparameter search via Ray Tune, optimizing F2 score, achieving macro recall of 0.574 (CV) and 0.571 on held-out test set, with diabetes class recall of 0.782 using custom clinical probability thresholds (Prediabetes >= 0.40, Diabetes >= 0.34).

- Built a full-stack platform (Next.js 14 + FastAPI) with offline-capable checkup history via IndexedDB, user authentication (PyJWT), a feedback loop API for continuous model improvement, and drift monitoring via Evidently AI; tracked all experiments in MLflow with DagsHub as remote registry.

- Integrated Google GenAI to generate personalized health recommendations from per-patient SHAP feature attributions, translating model confidence scores into actionable lifestyle and screening guidance; deployed frontend to Vercel and containerized backend to HuggingFace Spaces via Docker SDK.

Live: https://diawatch.vercel.app/   Code: https://github.com/JimRaph/DiaWatch

MediFact - Medical Retrieval-Augmented Generation (RAG) System

Python, FastAPI, ChromaDB, FastEmbed, LLaMA-3, AWS S3, Airflow, Docker, Modal, GitHub Actions, HuggingFace

- Architected a serverless RAG system on Modal (GPU inference) serving WHO factsheet data, with streaming responses via Server-Sent Events reducing TTFT significantly; optimized cold-start UX by pre-pinging the Modal container on page load and surfacing a live model status indicator bringing worst-case warm query latency to under 30s and simple queries under 5s.

- Engineered a multi-stage retrieval pipeline: Query Intent Classification -> query expansion -> semantic search (ChromaDB + BGE-small via FastEmbed) -> Cross-Encoder re-ranking -> token-aware context pruning; strict citation guardrails and safety prompting to prevent non-medical or harmful responses.

- Built an automated MLOps data pipeline using GitHub Actions (prod) and Airflow (dev) with AWS S3 for artifact storage & hash-based change detection to avoid redundant reprocessing of WHO datasets on bi-monthly ingestion runs.

Live: https://medifact.vercel.app/   Code: https://github.com/JimRaph/MediFact

SKILL

Languages & Frameworks
Python, TypeScript, FastAPI, Next.js, React, SQLAlchemy

ML & AI
CatBoost, XGBoost, Scikit-learn, SHAP, Ray Tune, FastEmbed, RAG, LLaMA-3, Google GenAI

Data & MLOps
Pandas, MLflow, Docker, AWS S3, GitHub Actions, Evidently, Airflow, DagsHub

Databases
PostgreSQL, IndexedDB, ChromaDB

Other
TailwindCSS, PyJWT, Framer Motion, Modal, HuggingFace Spaces

WORK EXPERIENCE

NYSC – Data Analysis Instructor Feb 2024 – Feb 2025
AIGE, Port Harcourt, Nigeria

- Delivered corporate data analysis training to 15+ staff across client organisations, building tailored curricula around each company's own datasets to ensure immediate workplace relevance.
- Led open-enrollment training to 200+ participants covering Python (Pandas, NumPy), Power BI, and Excel from data cleaning and transformation through to dashboard visualization.

Xorion Network - Full-Stack Developer June 2025 - Dec 2025
Wyoming, United States of America

- Built and deployed a production web application handling real-time cross-chain data (Substrate + Ethereum), inclucing portfolio dashboards with live state synchronisation between blockchain backends and the UI.
- Architected a bidirectional token bridging system with a custom Relayer implementation; refactored legacy codebase for maintainability under high-volume transaction load.

EDUCATION

University of Uyo, B.Eng. Computer Engineering (GPA: 4.0/5.0) October, 2023